

Transistor-level layout

Malte Schürks

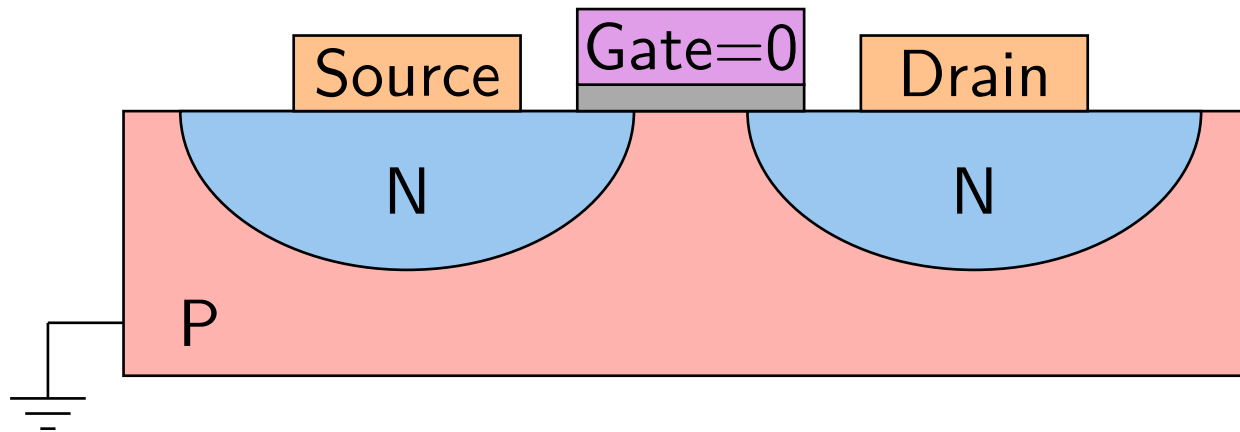
University of Bonn

September 21, 2026

What is a FET?

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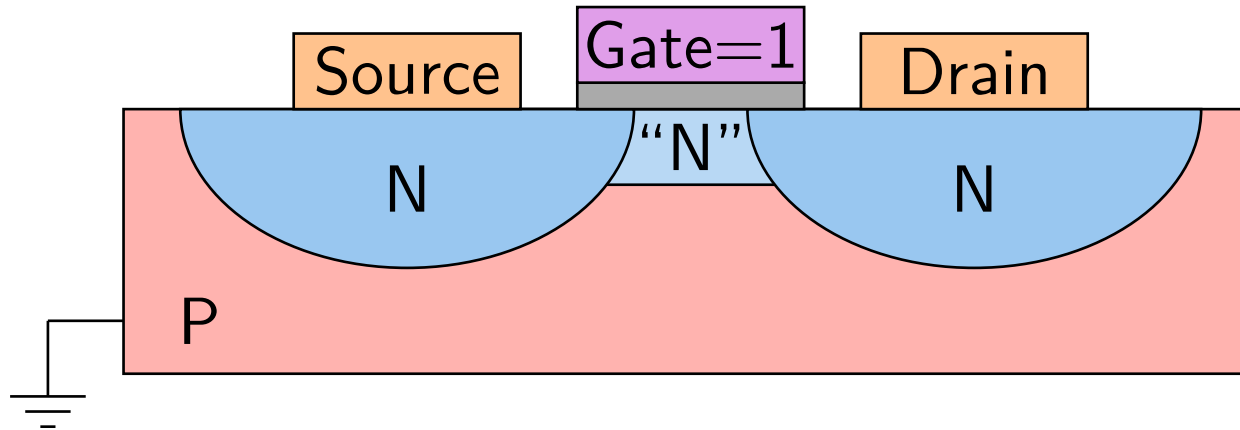
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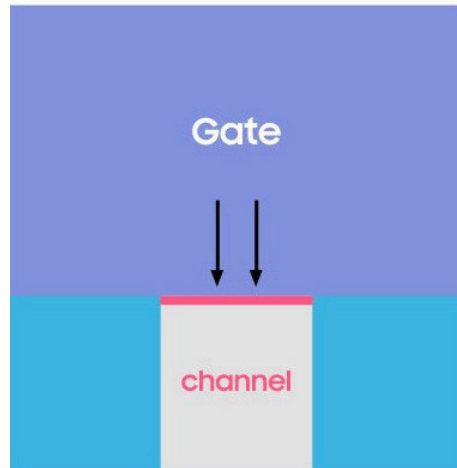
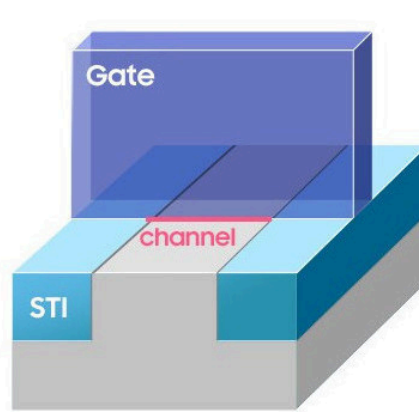
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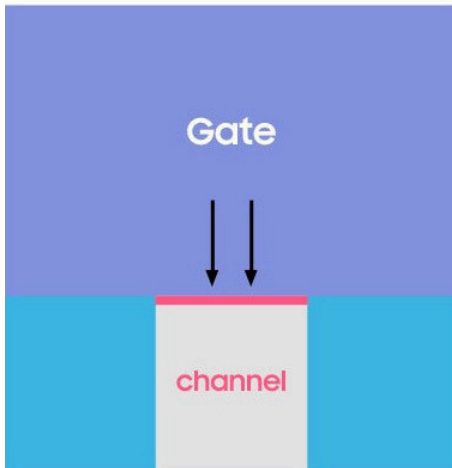
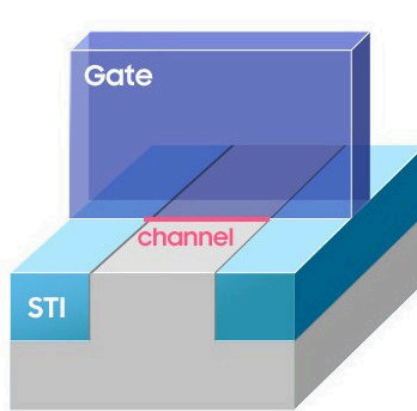
FinFET and Gate-All-Around FETs

Planar

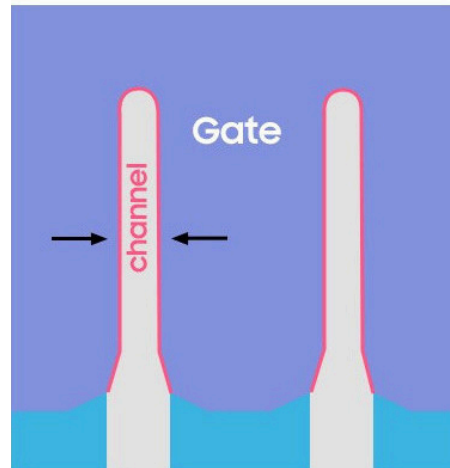
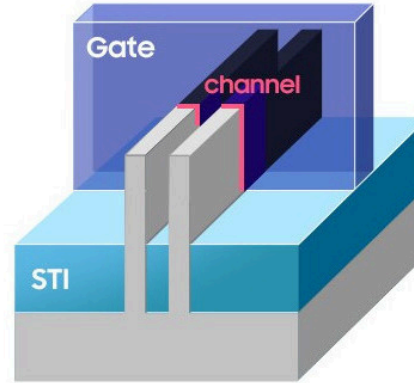


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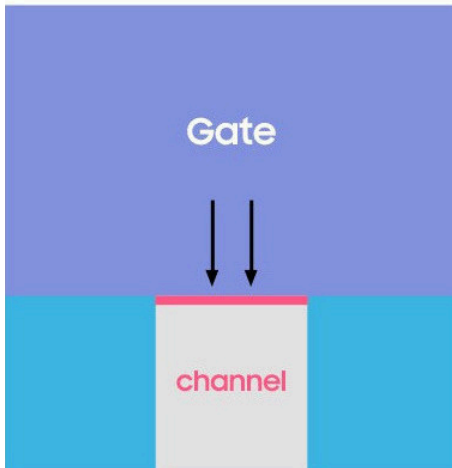
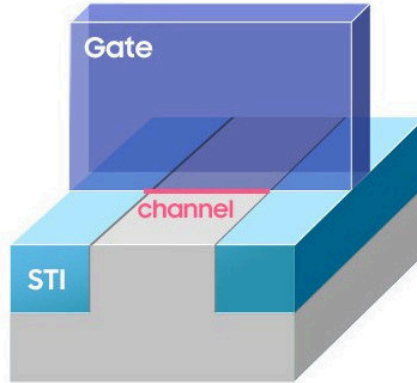


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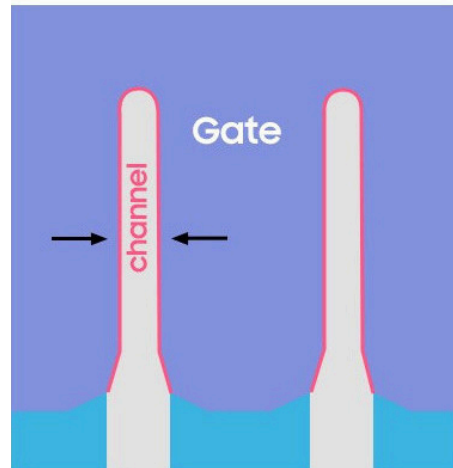
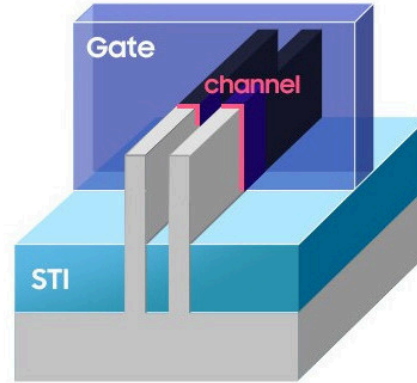


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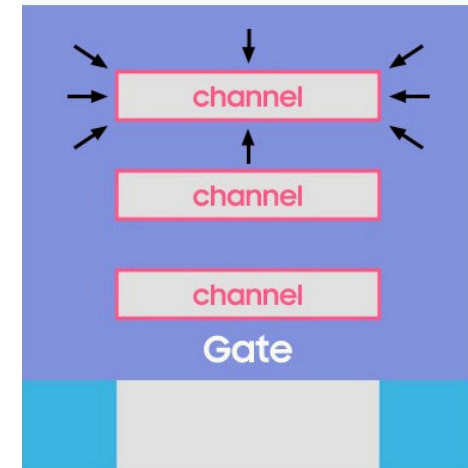
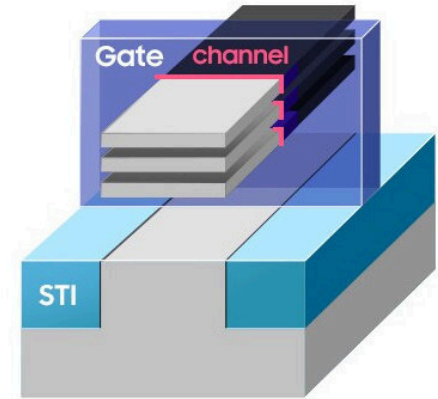
Planar



FinFET



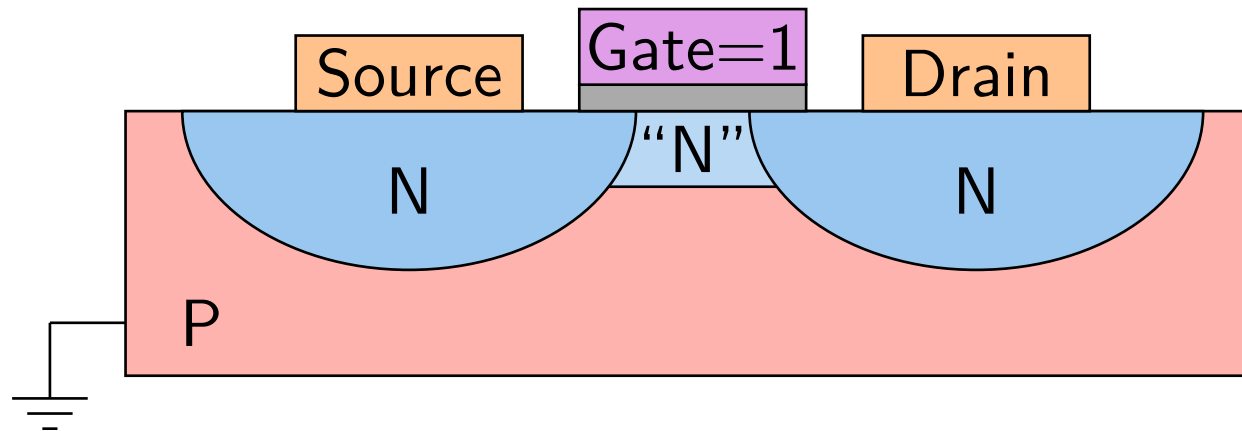
Gate-All-Around



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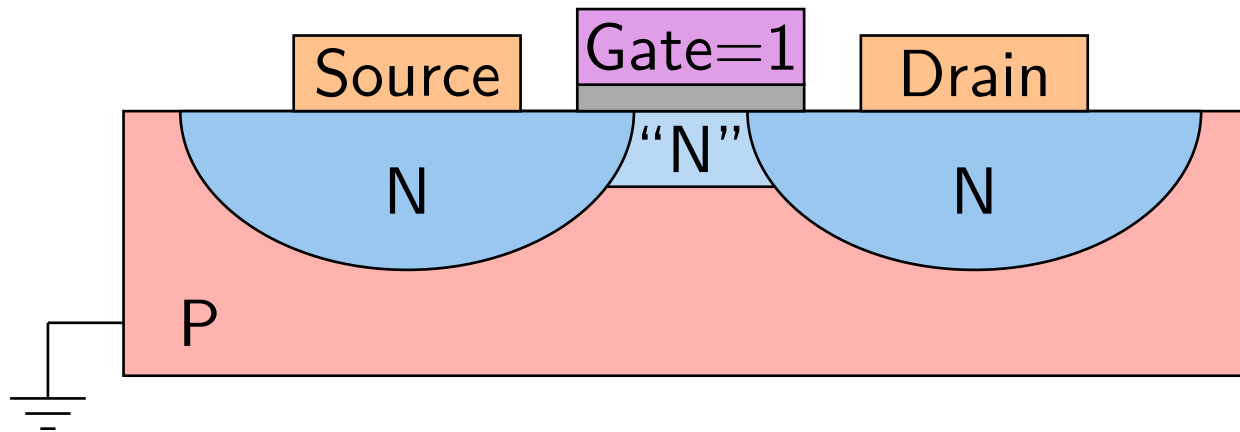
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 - **N**(-channel) FET: Conducts if gate at logic 1



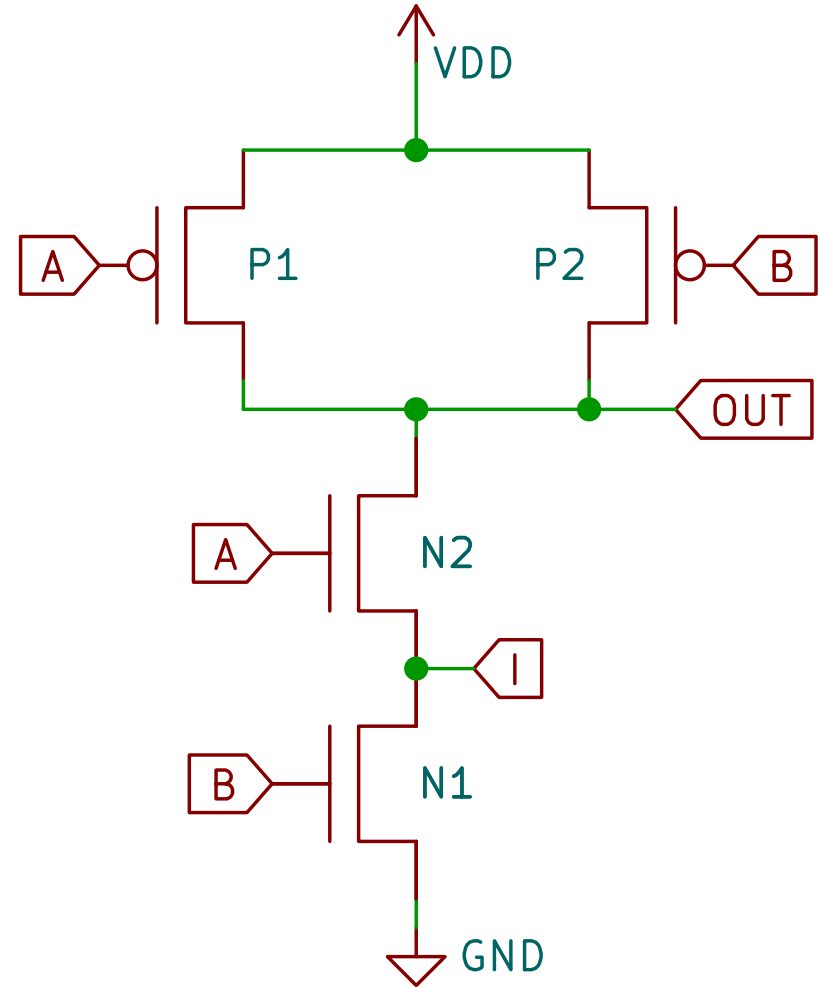
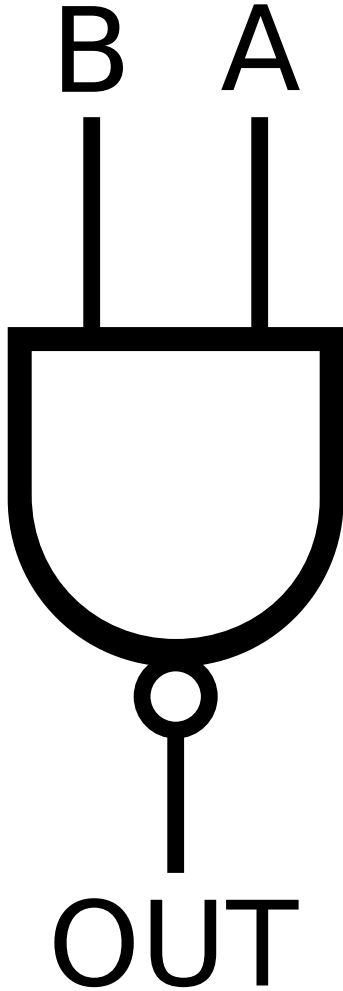
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 - **P**(-channel) FET: Conducts if gate at logic 0



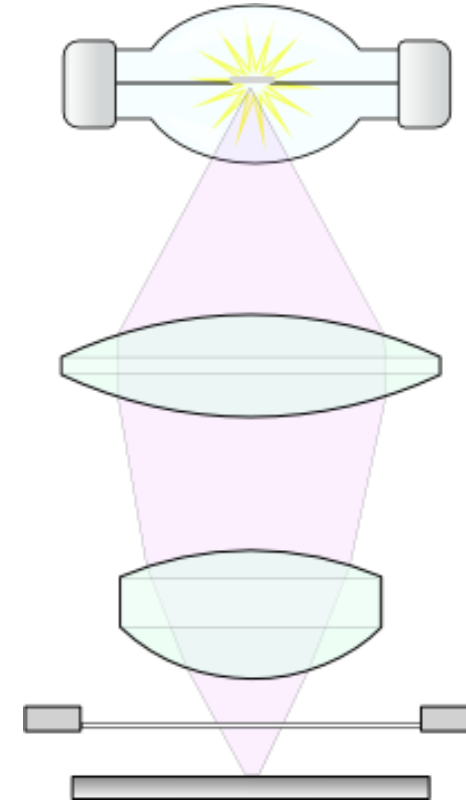
Building logic from FETs



Lithography and design rules

Important manufacturing step:

Photolithography

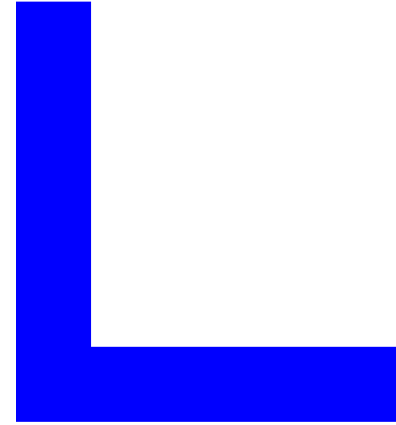


Lithography and design rules

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Lithography and design rules

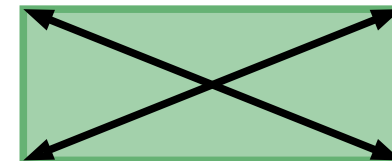
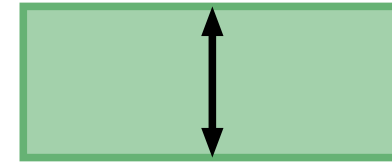
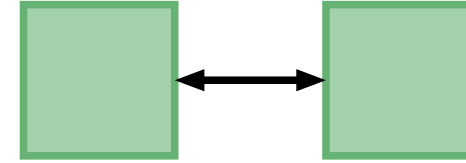
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Fix: Forbid problematic patterns

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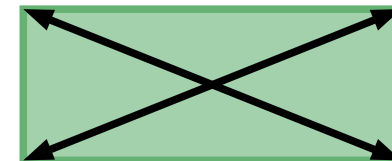
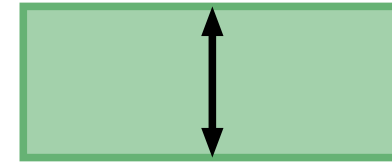
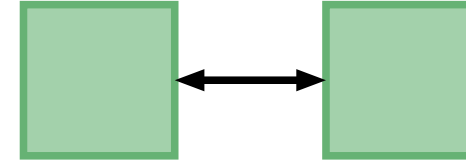
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Common rules:

Min spacing, width, area, ...

Overall: ~ 1300 pages for 2nm



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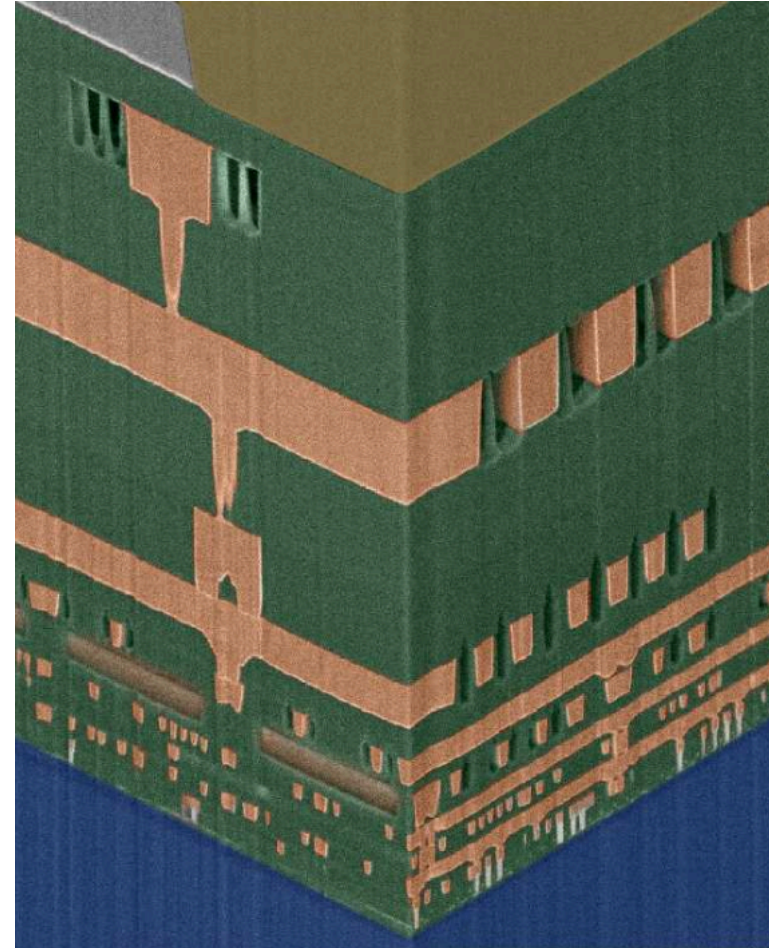
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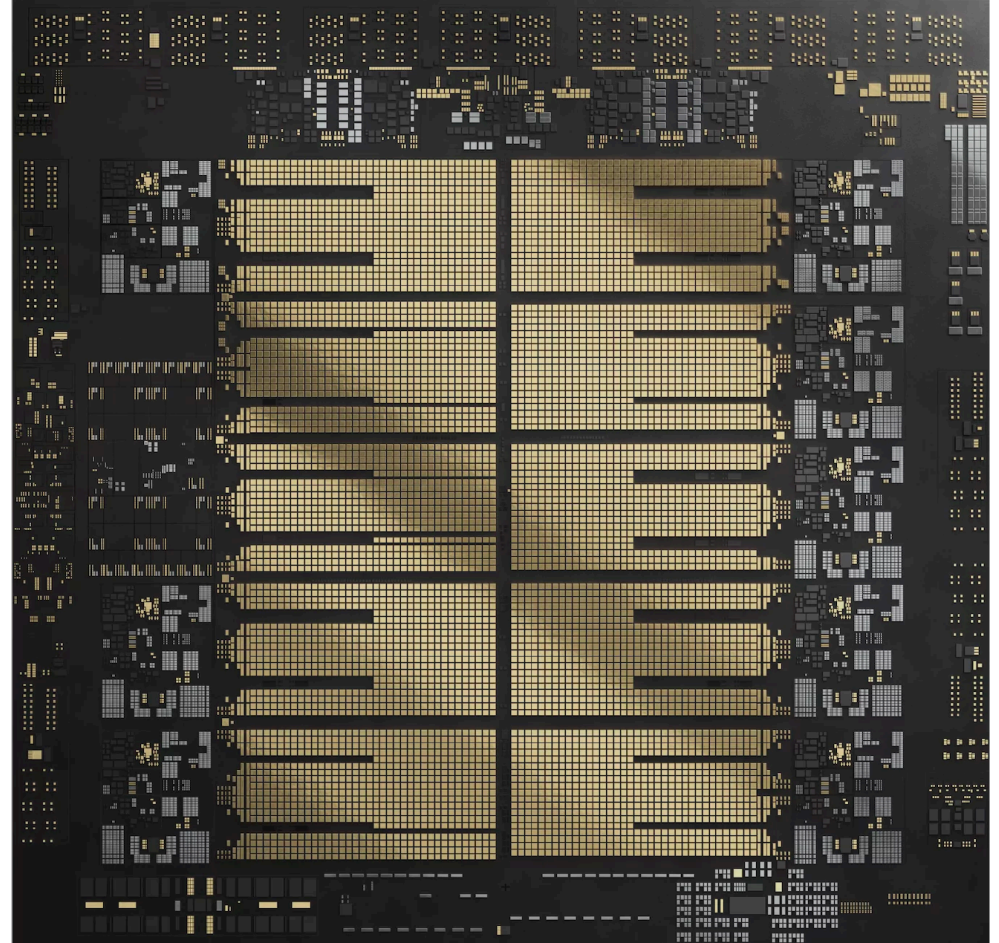
Overall: ~ 1300 pages for 2nm

Higher density $\Rightarrow$ more rules



Motivation for standard cells

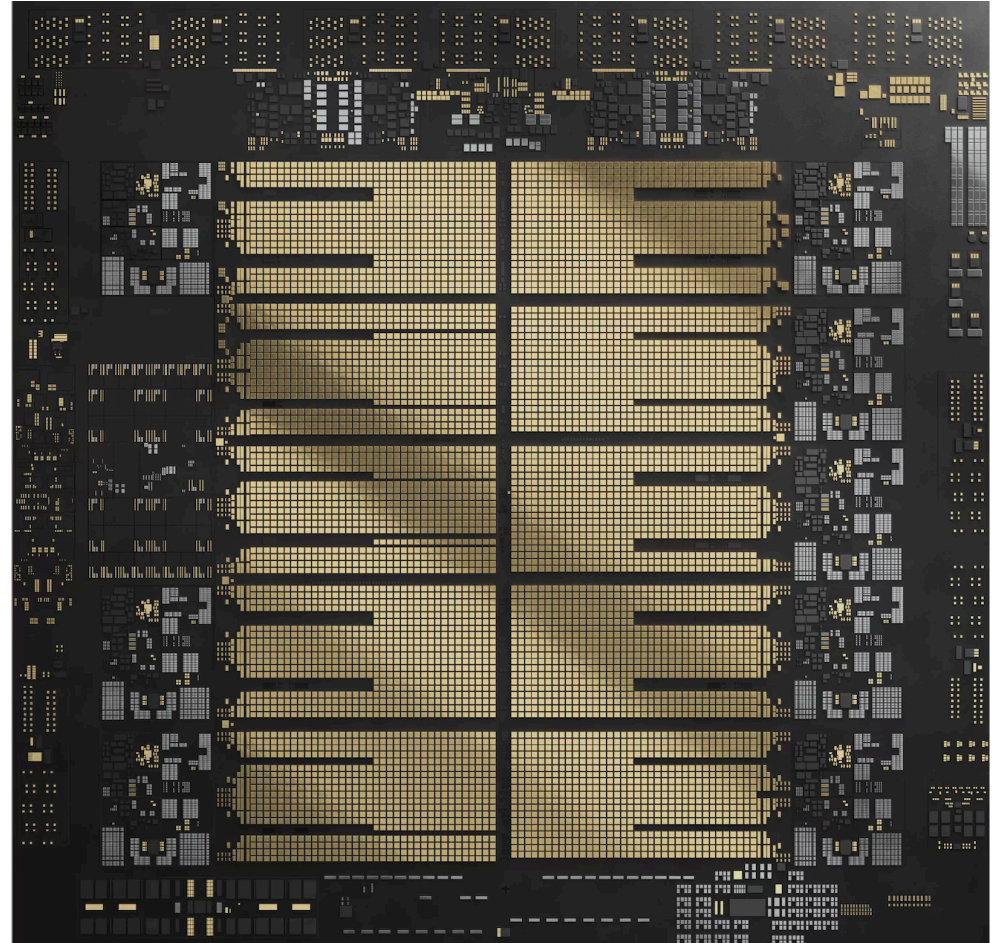
- Modern chips: $> 40 \cdot 10^9$ FETs
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⇒ “Hide” FETs inside **cells**:
Logic, latches, ...



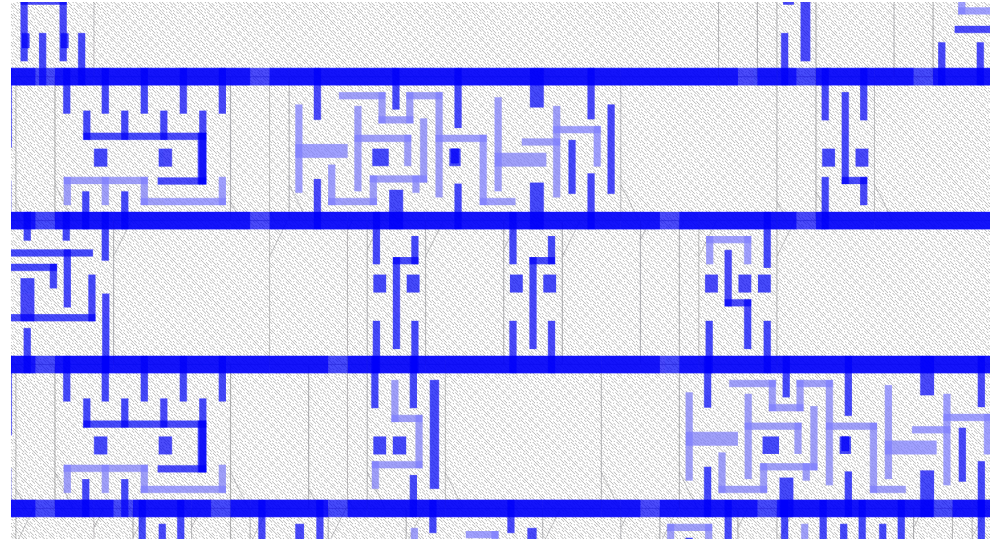
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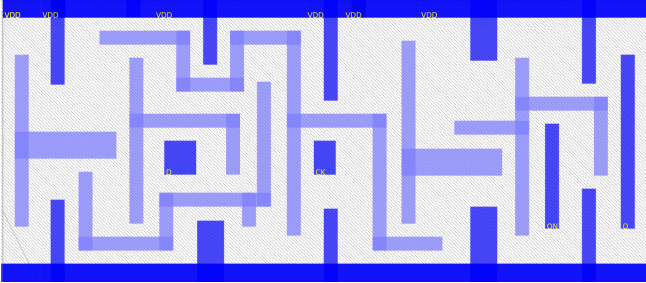
⇒ “Hide” FETs inside **cells**:
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Advantages:

- Smaller instance sizes
- Independent building blocks
- Easier timing analysis
- Reusable across chips



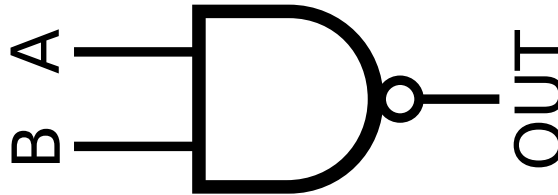
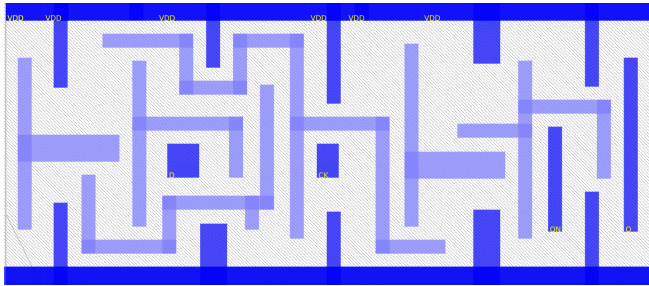
What defines a cell?



Geometric properties

- Size
- Possible positions
- Pin shapes
- Blockages

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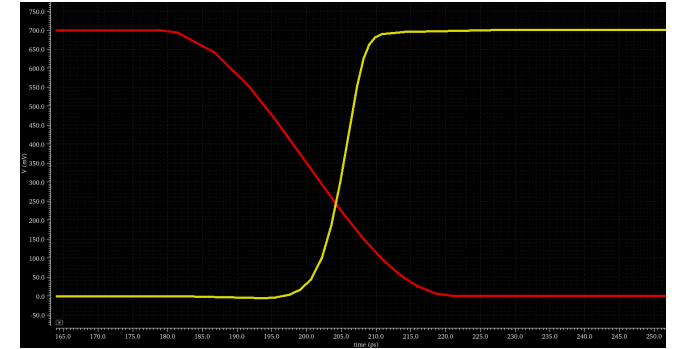
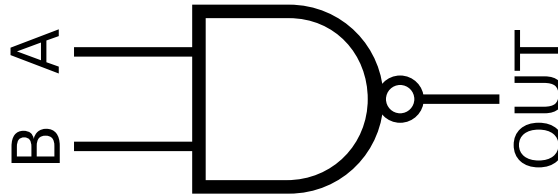
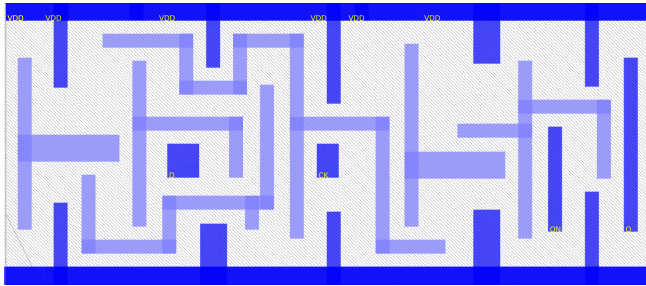
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- Potentially including state

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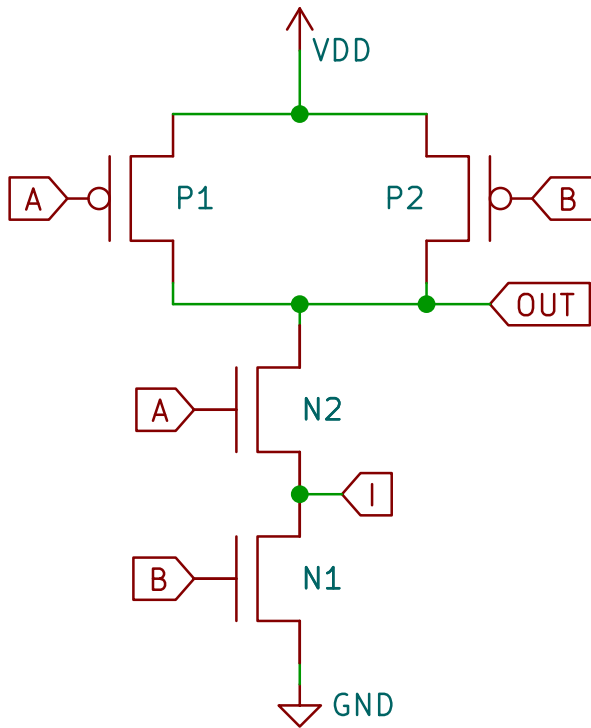
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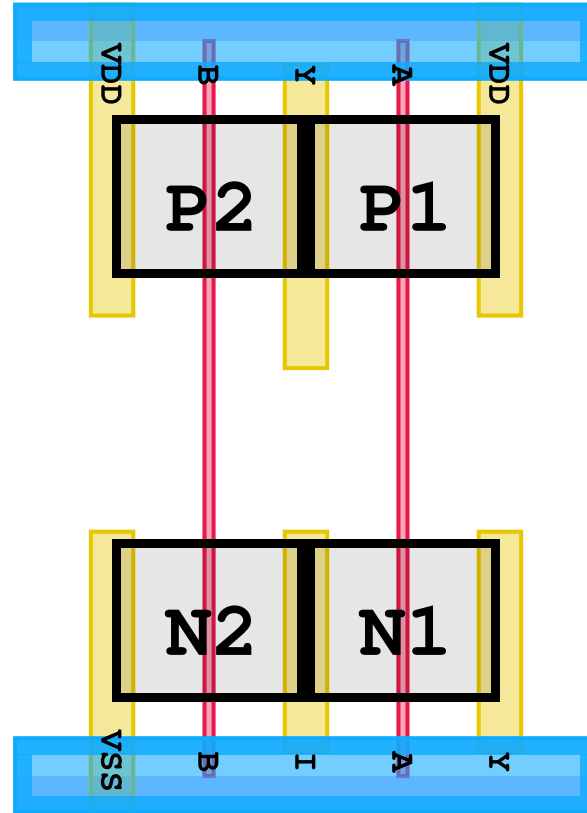
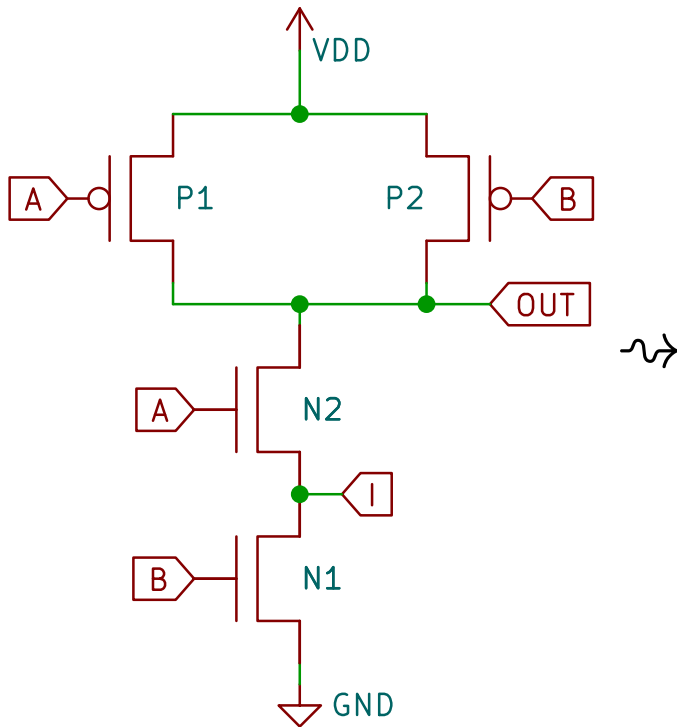
Timing and power

- Delays
- Static power
- Dynamic power

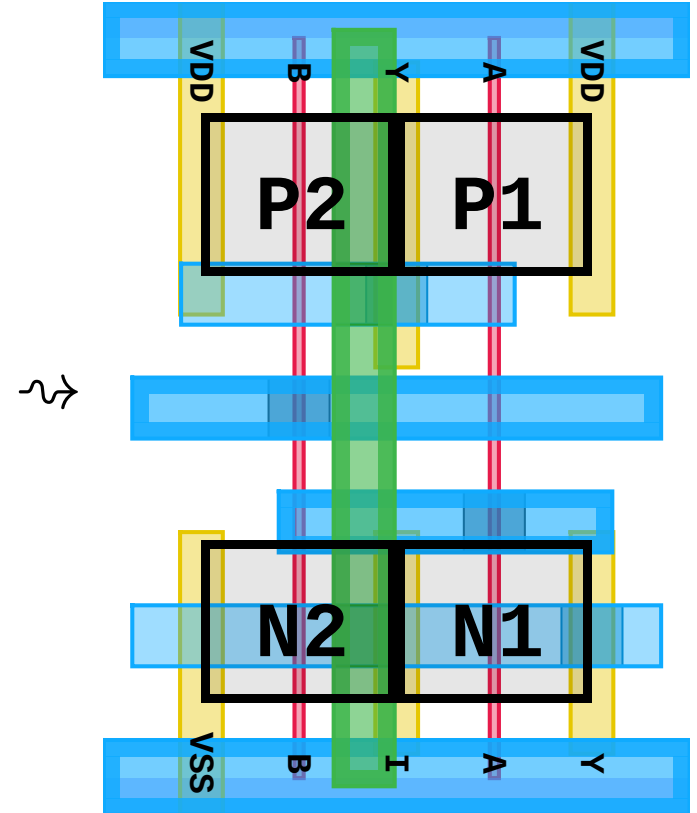
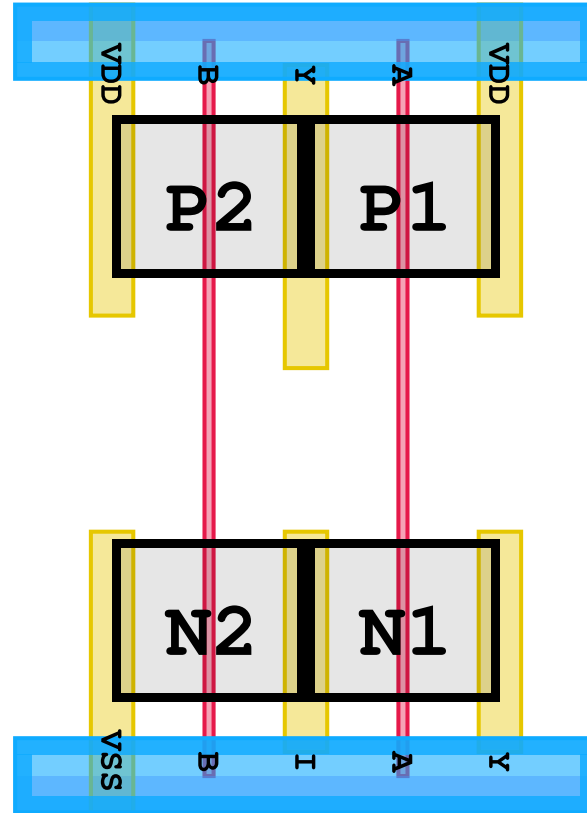
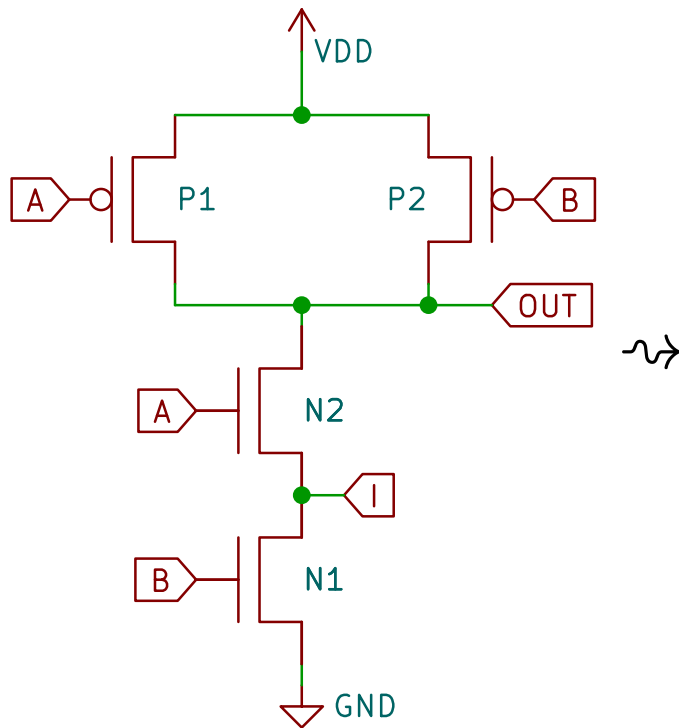
Building a cell



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Exact placement by Branch&Bound

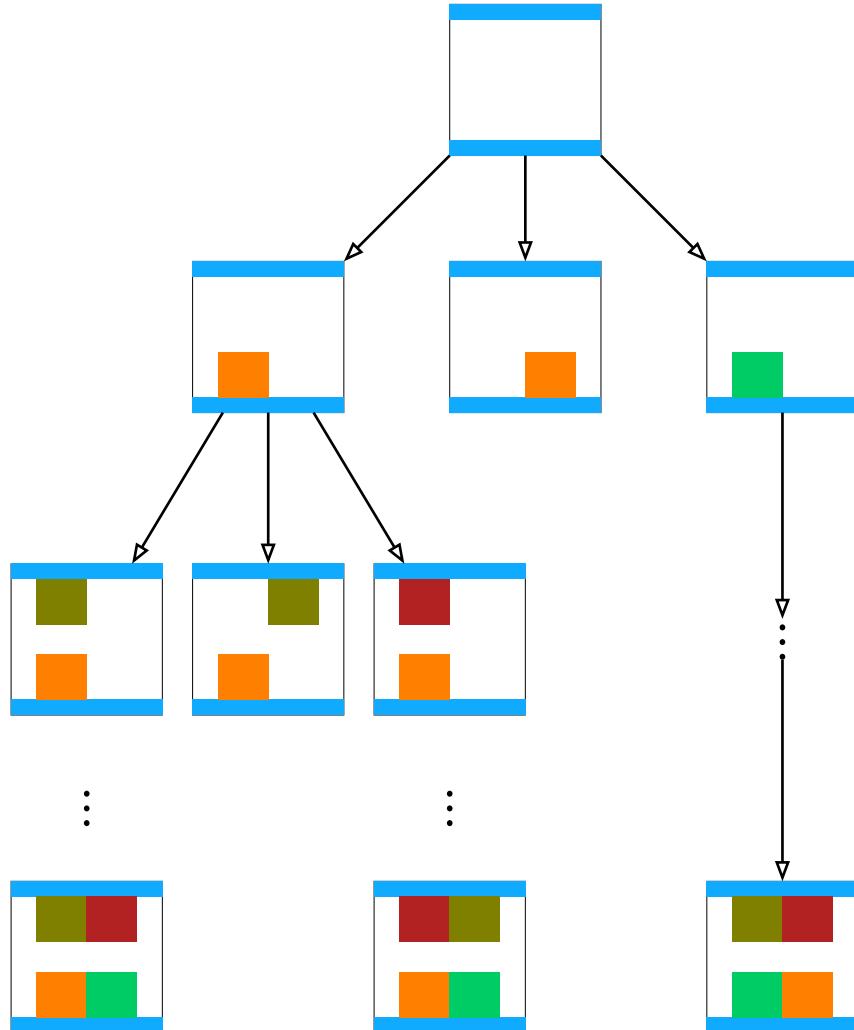
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Exact placement by Branch&Bound

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Algorithmic structure:

- Place FETs one at a time

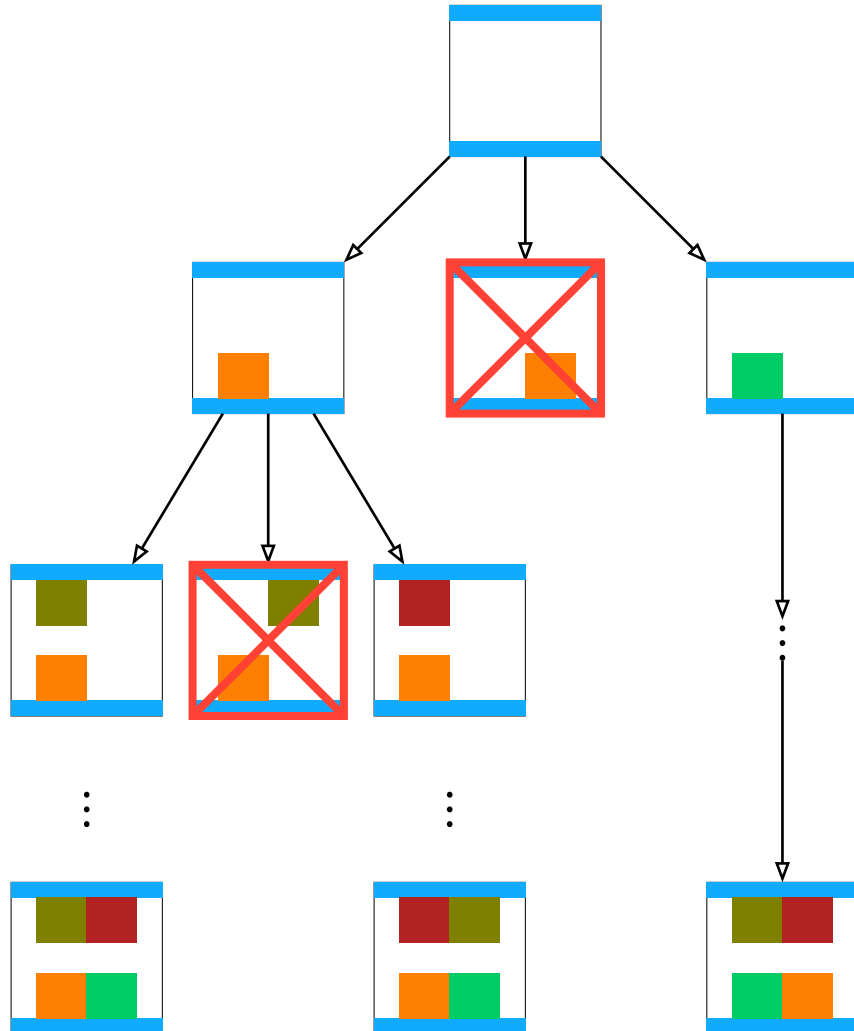


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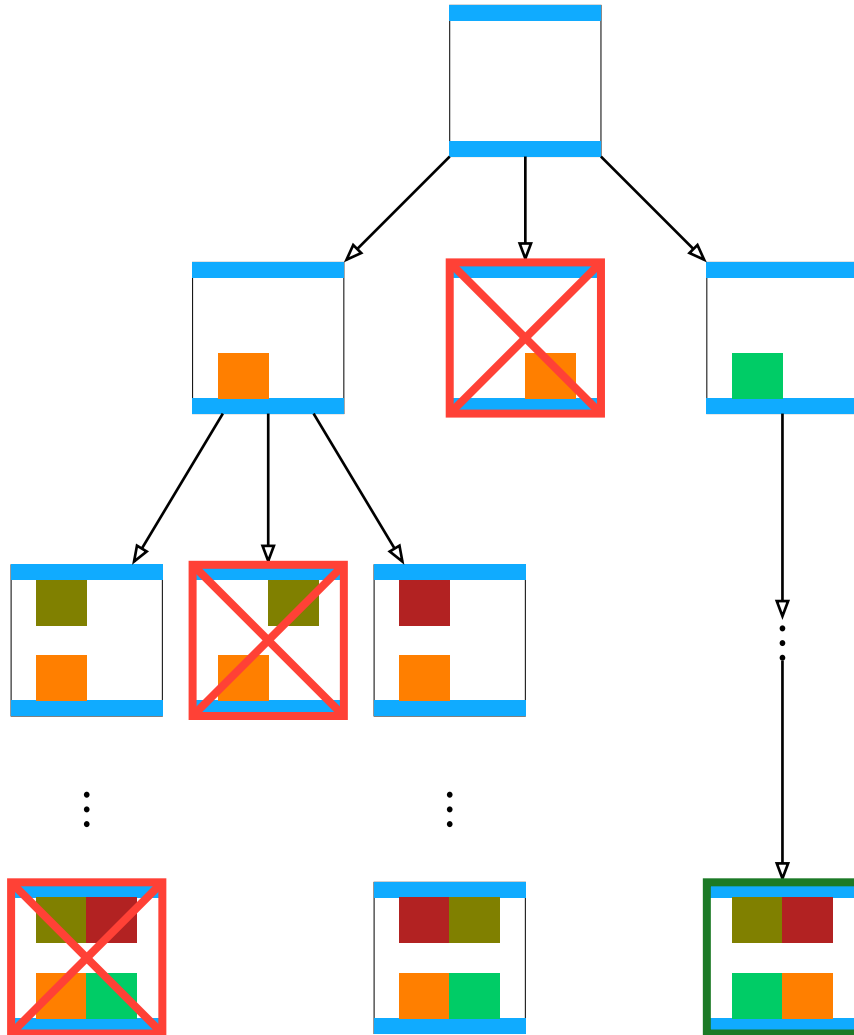


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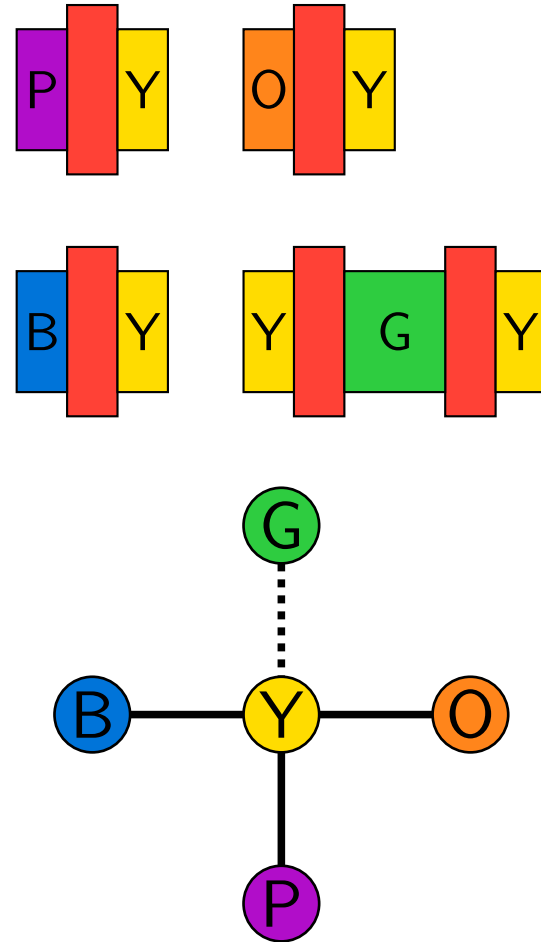
- Place FETs one at a time
- Prune useless branches early
- Check routability of full placements
- Find best remaining



Pruning techniques

Width bounds:

- Is there space for all FETs?
- Remove many useless branches
- Never prunes full nodes



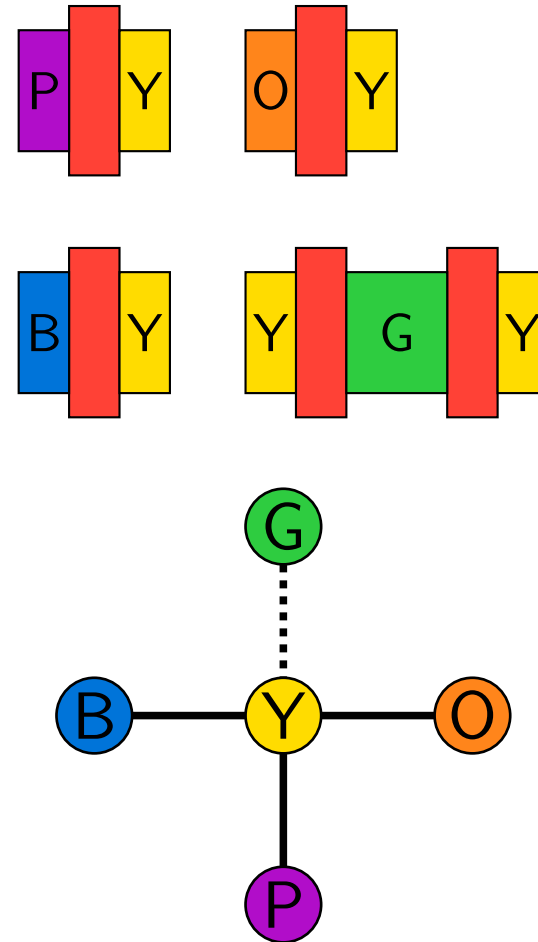
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Routing checks:

- Basic checks on partial nodes
- Prunes unroutable full nodes
- Only consider FEOL shapes
 - I.e. shapes “close to FETs”

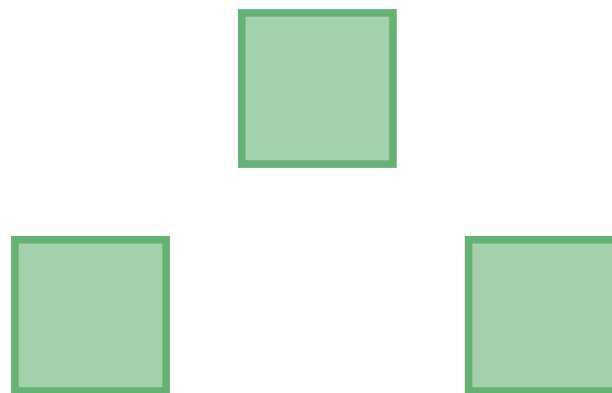


ILP-based routing

“Direct” routing algorithms:

- Usually heuristic
- Tricky to extend

Design rule: 3 incompatible vias



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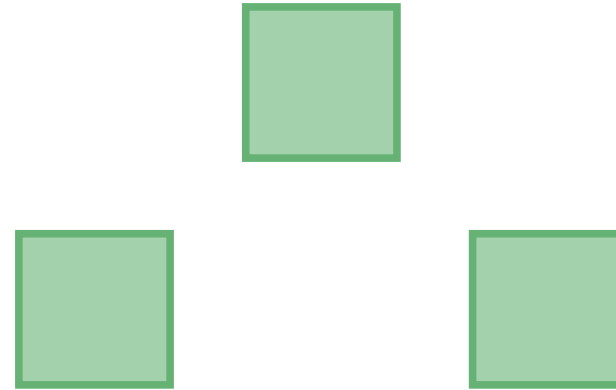
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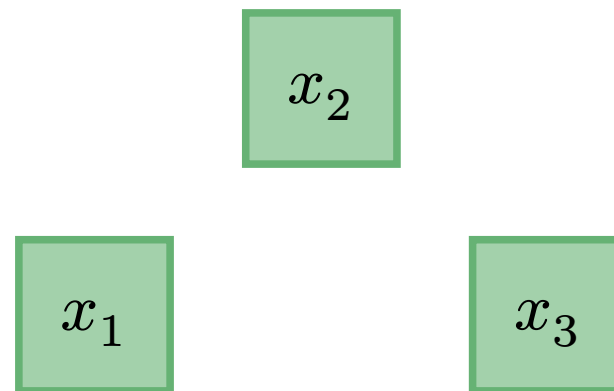
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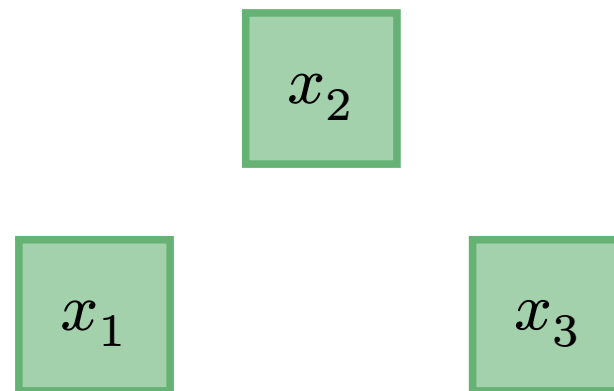
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ILP constraint: $x_1 + x_2 + x_3 \leq 2$

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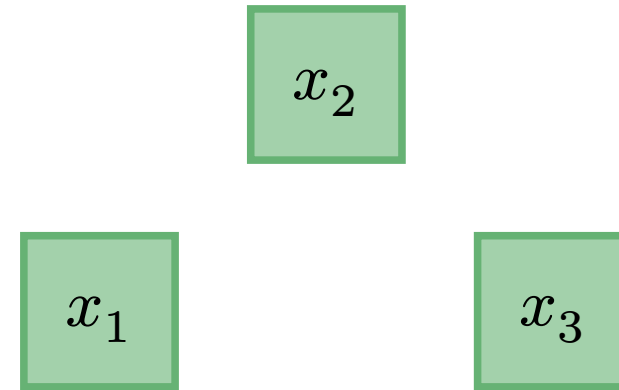
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Connectivity:

- Steiner tree per net
- Multicommodity flow relaxation

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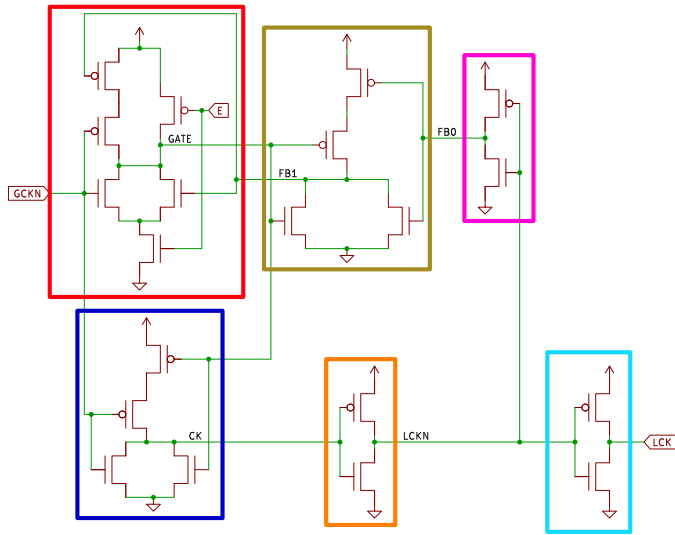
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Further concepts in BonnCell

Placing larger cells

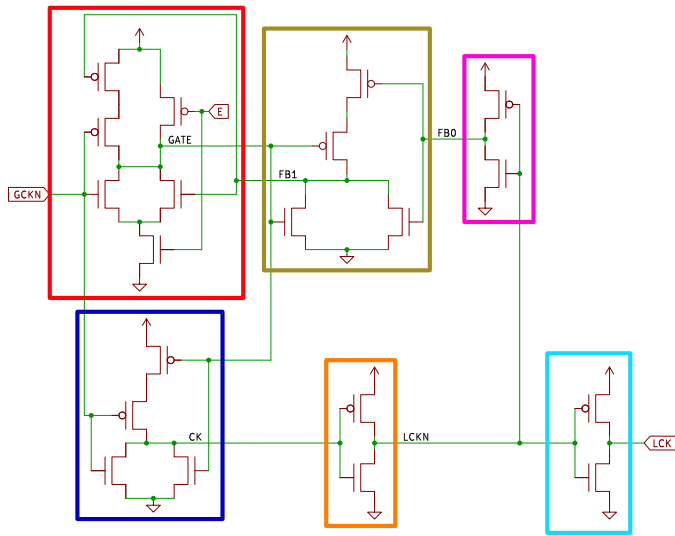
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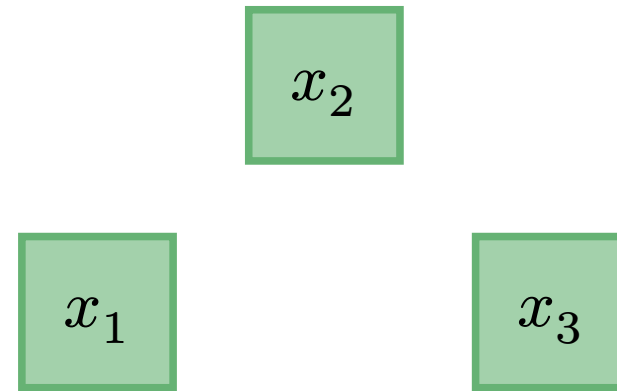
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Faster routing checks

- Bottleneck during placement
- Transform ILP to SAT
- Extension: Decision-guided SAT



$$x_1 + x_2 + x_3 \leq 2$$
$$\rightsquigarrow \neg x_1 \vee \neg x_2 \vee \neg x_3$$